

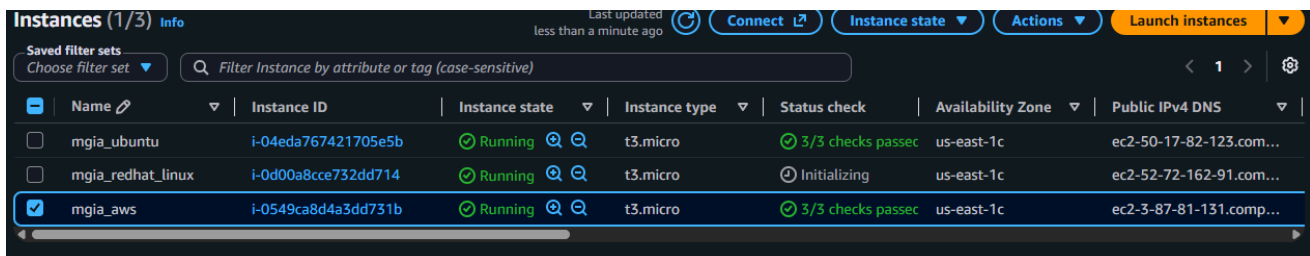
Assignment 3 - EC2 and EFS

Objective: Create one Amazon Elastic File System (EFS) and connect it to three different EC2 instances, with each instance using a different Linux operating system: Ubuntu, Red Hat Linux, and Amazon Linux.

1. EC2 Instances with Different Operating Systems

First, three EC2 instances were created in the AWS environment. The instances were configured with different operating systems so that the same EFS could be tested from multiple Linux distributions.

- Ubuntu
- Red Hat Linux
- Amazon Linux



Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
mgia_ubuntu	i-04eda767421705e5b	Running	t3.micro	3/3 checks passed	us-east-1c	ec2-50-17-82-123.com...
mgia_redhat_linux	i-0d00a8cce732dd714	Running	t3.micro	Initializing	us-east-1c	ec2-52-72-162-91.com...
mgia_aws	i-0549ca8d4a3dd731b	Running	t3.micro	3/3 checks passed	us-east-1c	ec2-3-87-81-131.comp...

Figure 1. Three EC2 instances created with different operating systems.

2. Connect to the EC2 Instances

The three EC2 instances were accessed using the AWS EC2 Instance Connect terminal. The screenshot below shows the instances being accessed and the installation of the NFS/EFS client utilities required for mounting the EFS file system.

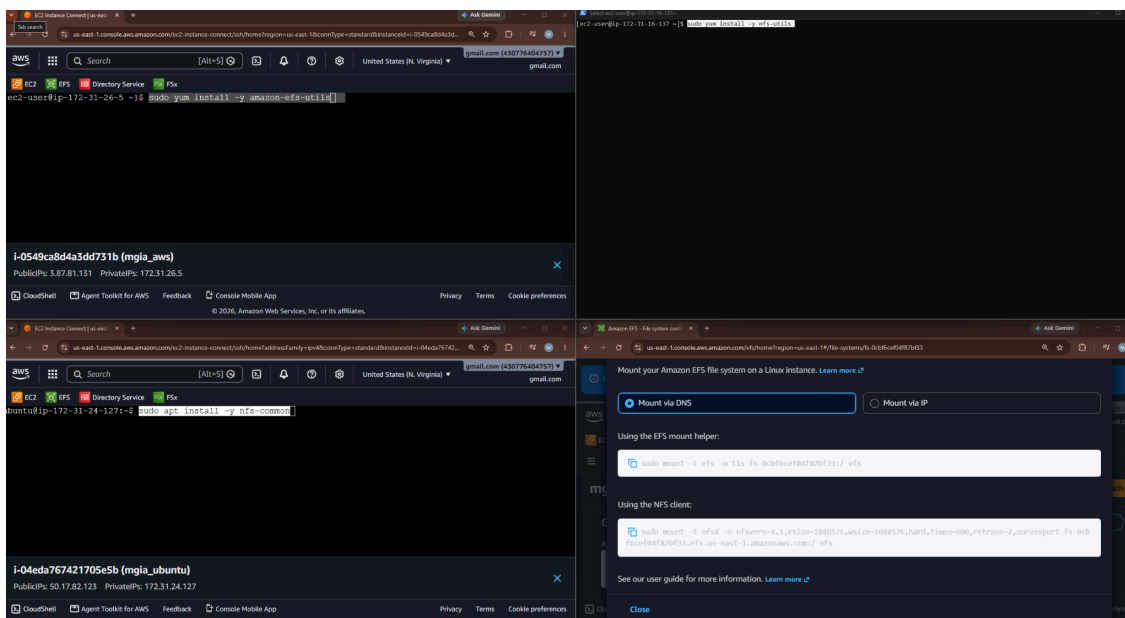


Figure 2. Connecting to the EC2 instances and installing the required NFS/EFS utilities.

Typical package installation commands used:

Ubuntu: `sudo apt update && sudo apt install -y nfs-common`

Red Hat: `sudo yum install -y nfs-utils`

Amazon Linux: `sudo yum install -y amazon-efs-utils`

3. Create the EFS Mount Directory

A directory named `efs` was created on the EC2 instances. This directory acts as the local mount point through which the EFS file system can be accessed.

`sudo mkdir -p ~/efs`

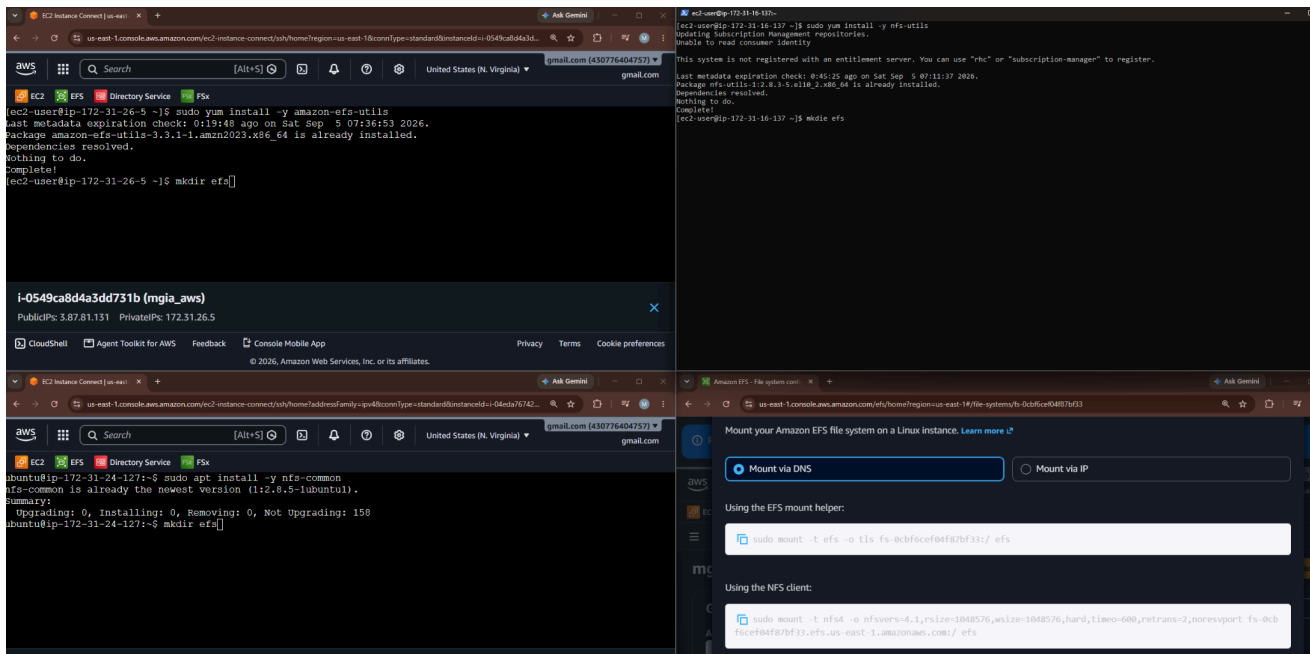


Figure 3. Creating the local directory used as the EFS mount point.

4. Create the Amazon EFS File System

Amazon EFS was opened from the AWS Management Console and a new file system was created. The EFS was created in the same VPC as the EC2 instances so that the instances could communicate with the EFS mount targets.

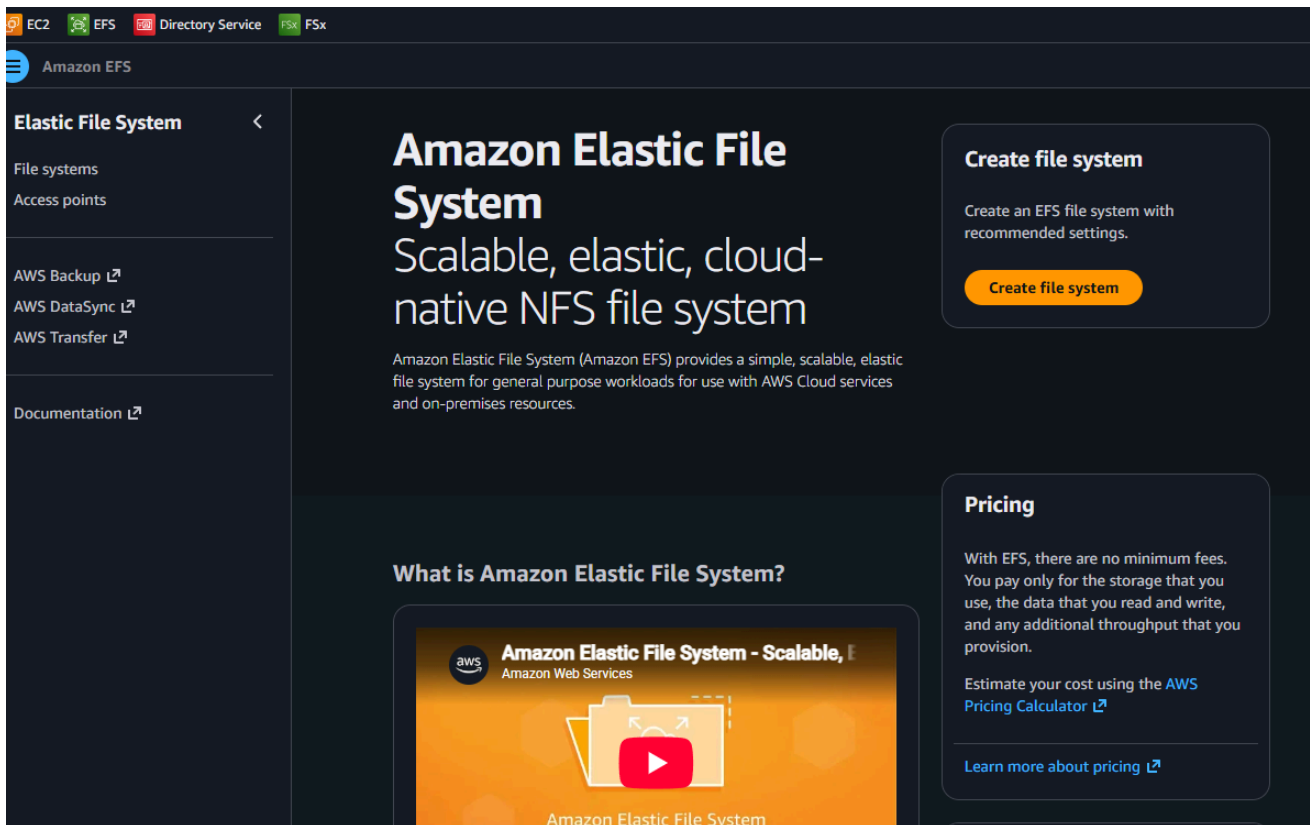


Figure 4. EFS file system created in the AWS console.

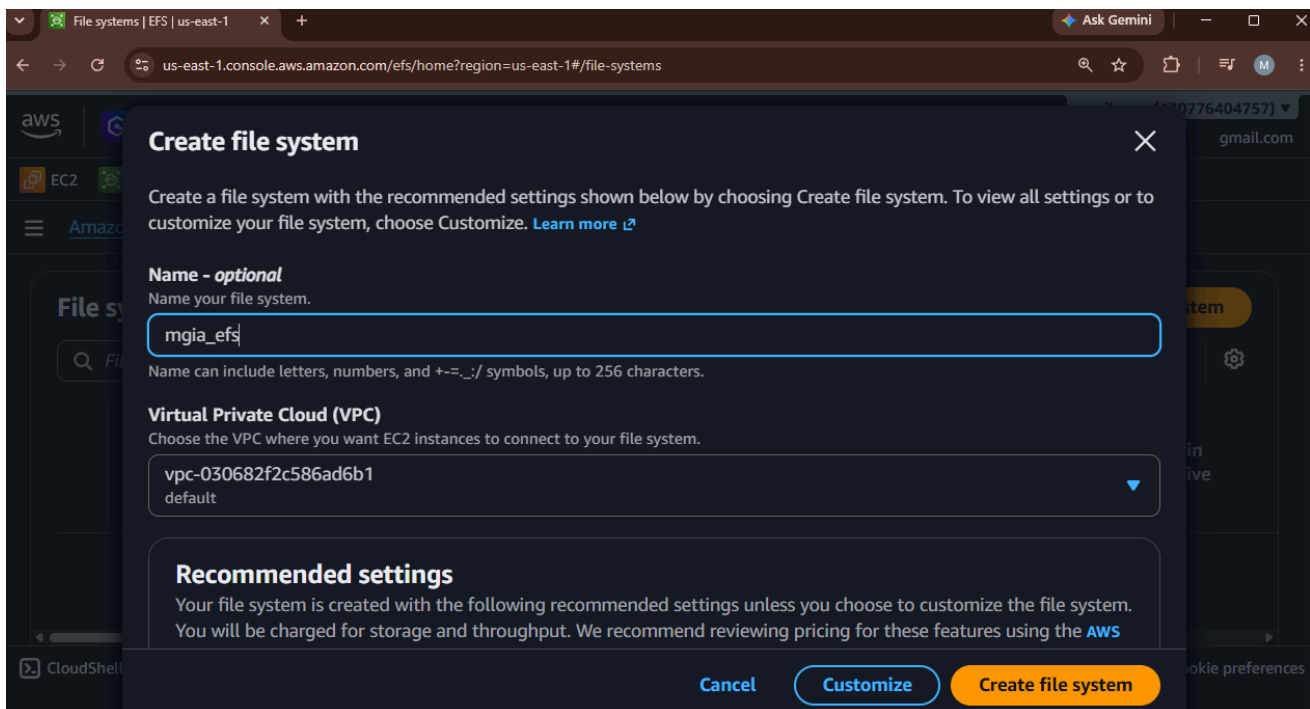


Figure 5. EFS creation screen showing the file-system name and VPC selection.

5. Mount the EFS on the EC2 Instances

After creating the EFS, the AWS console provides the commands required to mount the file system. The NFS client was used to connect the Linux instances to EFS.

```
sudo mount -t nfs4 -o nfsvers=4.1,rsize=1048576,wsize=1048576,hard,timeo=600,retrans=2,noresvport  
<EFS-DNS-NAME>:/ ~/efs
```

Replace <EFS-DNS-NAME> with the DNS name provided by the EFS console. The same EFS file system can then be mounted on each of the three EC2 instances.

```
sudo mount -t nfs4 -o nfsvers=4.1,rsize=1048576,wsize=1048576,hard,timeo=600,retrans=2,noresvport  
fs-0cbf6cef04f87bf33.efs.us-east-1.amazonaws.com:/ ~/efs
```

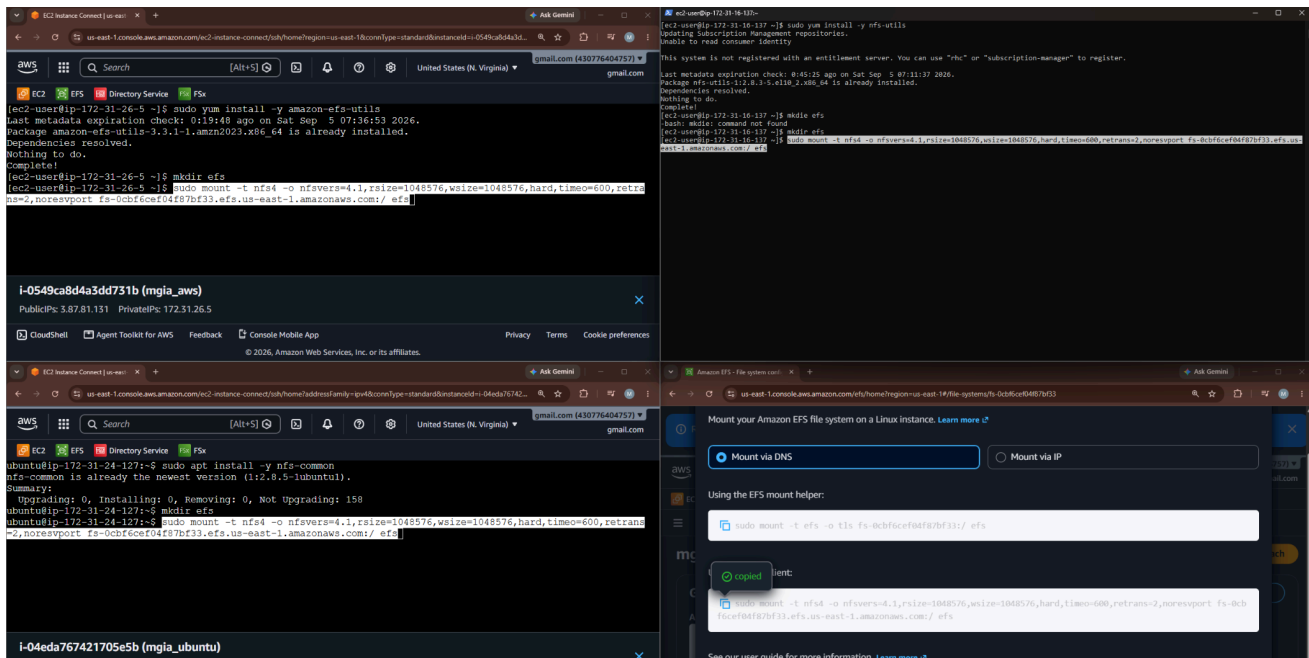


Figure 6. EFS mount instructions and the NFS mount command.

6. Verify the EFS Connection

After mounting, the EFS directory can be checked from the terminal. The presence of the efs directory confirms that the mount point has been created. The mounted file system can also be verified using df -h.

Verification commands:

ls

df -h ~/efs

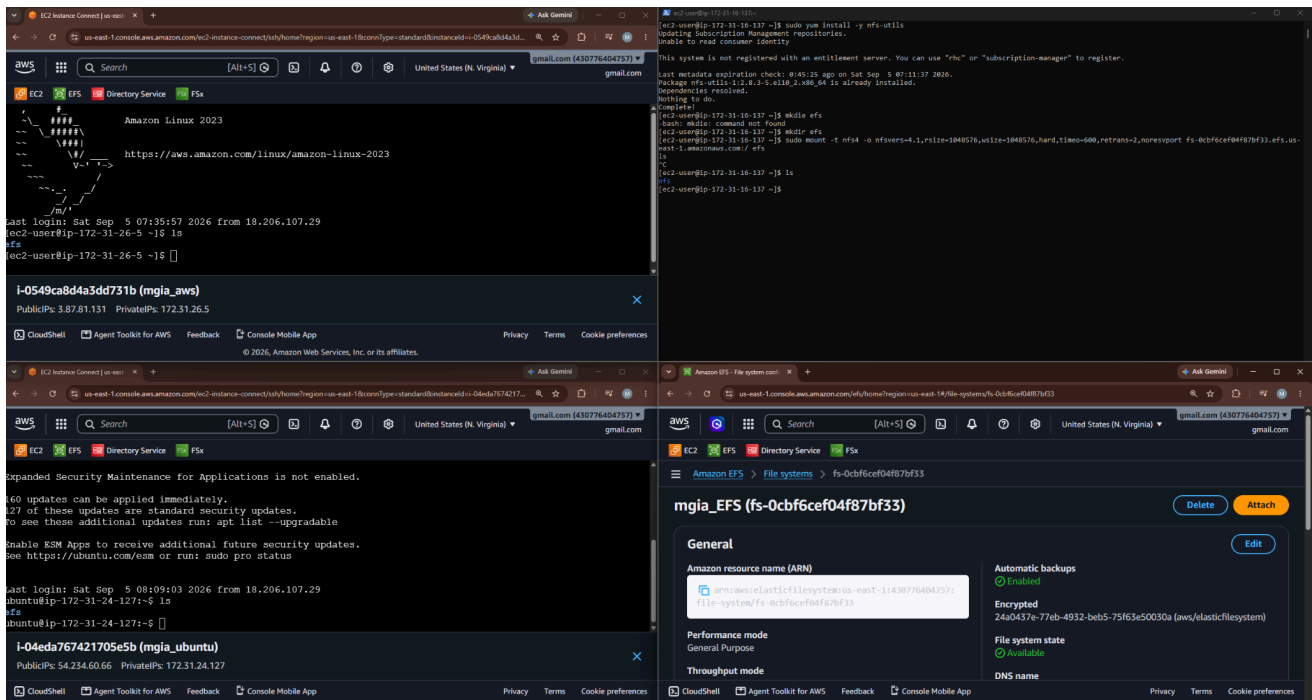


Figure 7. Verifying the EFS directory from the EC2 terminal.

7. Final Result

The task was completed by creating a single Amazon EFS file system and connecting it to three EC2 instances running different operating systems. Because EFS is a shared network file system, the same files can be accessed from all connected instances.

EC2 Instance	Operating System	EFS Mount Point
EC2-1	Ubuntu	~/efs
EC2-2	Red Hat Linux	~/efs
EC2-3	Amazon Linux	~/efs